

Sreevatsa RAVIKANTH, MSc in Bioinformatics

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Portfolio



Employment History

Technical experience

- Apr 2026 – Jul 2026 ■ **Full Stack Engineer**, Districom, Montigny-lès-Cormeilles, France.
- Development of an mobile application for retail stock management for Sony France with Dart/Flutter
- Apr 2025 – Feb 2026 ■ **Webmaster**, nutritionDay worldwide, Villejuif, France.
- Creation and management of the French branch website for the international nutritionDay study
- Adaptation of an international clinical research project to the French healthcare context
- Technical maintenance, user support, and coordination of national participation in a global epidemiological study
- Mar 2025 – Oct 2025 ■ **Biostatistician**, Gustave Roussy, Villejuif, France.
- Contribution to scientific writing (second author of a manuscript under review)
- Statistical study on micronutrient deficiencies in head and neck cancer patients
- Construction and structuring of clinical databases for predictive modeling
- Univariate + multivariate and LASSO logistic regressions
- Apr 2024 – Sep 2024 ■ **Biostatistician**, Gustave Roussy, Chevilly-Larue, France.
- Contribution to a scientific manuscript under review
- Data preprocessing and statistical analysis of 204 oncology patients
- Clinical data structuring for research in oncology nutrition
- Calculation of prevalences and subgroup comparisons using Boschloo's exact tests
- Identification of 6 significant micronutrients related to inflammation through Spearman correlations and symbolic regression modeling (95% CI)
- Jan 2023 – Feb 2023 ■ **Process Optimization Consultant**, D-MUTE NETWORK, Saint-Maurice, France.
- Design and development of APIs for enterprise resource management
- Integration of Microsoft 365 licensing data into SharePoint systems

Other experience

- May 2018 – Present ■ **Elderly Care Assistant**, URSSAF Île-de-France, Joinville-le-Pont, France.
- Apr 2026 – Present ■ **Product Promoter**, Sony France / Districom, Montigny-lès-Cormeilles, France.
- Dec 2021 – Aug 2022 ■ **Sales Advisor**, Xiaomi France, Paris, France.
- Feb 2016 – Apr 2016 ■ **Animator**, Youth and Archive Department, City of Joinville-le-Pont, France.

Education

- 2020 – 2025  **Engineering Degree (MSc equivalent), École Centrale d'Électronique Paris**, in Computer Science & Electronics, specialization Health & Technology.
Master's Thesis: *Rapport de stage de fin d'études*.
International exchanges: *University of Verona (2024)*, *OMNES London School (2022)*.
- 2017 – 2020  **Scientific Baccalaureate, Lycée Teilhard de Chardin**, Specialization Mathematics & Earth and Life Sciences (SVT), with high honours.

Research Publications

Journal Articles

- 1 M. Guinhut et al., "Prevalence of micronutrient deficiencies in 465 patients with head and neck cancers (hnc) admitted to a rehabilitation unit," *Clinical Nutrition ESPEN*, 2025, Manuscript under review.
- 2 A. Konesalingam, S. Ravikanth, G. M. Machado, F. Belbachir, and A. Jaouadi, "Stress detection with physiological signal from wrist-worn devices," 2025, Unpublished manuscript.

Research Projects

- 2023 – 2025  **Stress Detection Using Machine Learning**, LyRIDS, Paris, France
Co-lead author of a unpublished scientific manuscript
- Harmonization of multimodal physiological datasets (PPG, TEMP, EDA, ACC)
- Advanced signal preprocessing: filtering, temporal and frequency feature engineering, dimensionality reduction
- Analysis of multimodality impact and windowing strategies
- Training and evaluation of 12 machine learning models using LOSO-CV
- Bayesian hyperparameter optimization
- Best performance: F1-score of 81.9% (ternary classification)
- Model interpretability using SHAP values
- Development of reproducible ML pipelines on Linux systems
- 2024  **Autonomous Drone Navigation Using Reinforcement Learning**, LyRIDS, ECE Paris, France
- Built a custom dataset of 180 subjects, with face segmentation to isolate facial regions from raw images
- Applied random rotation for data augmentation to artificially increase the dataset size
- Retrained a pretrained ResNet101 convolutional neural network under MATLAB
- Achieved a 90% correct detection rate

Research Projects (continued)

- **Single-Cell RNA-seq Analysis of Cancer Samples**, University of Verona, Italy
6-month MSc research project
- Built a reproducible scRNA-seq analysis pipeline for 10x Genomics datasets using Seurat in R and R Markdown
 - Performed quality control, normalization, Highly Variable Gene Selection, PCA, UMAP and graph-based clustering
 - Improved data quality through automated and supervised doublet detection using DoubletFinder and lineage-specific markers
 - Annotated cell populations using SingleR, CHETAH and canonical marker genes
 - Identified cluster-specific marker genes and characterized immune and stromal populations within the NSCLC tumor microenvironment
 - Investigated cell-cell communication with CellChat and lineage trajectories using Sling-shot
- 2023 ■ **Face Detection Using Transfer Learning**, OMNES London School, United Kingdom
- Retrained a pretrained ResNet convolutional neural network under MATLAB
 - Achieved a 90% correct detection rate
- 2020 – 2022 ■ **ECE CUP**, LyRIDS, Paris, France
Semi-finals, ranking among the top teams out of more than 300 competing teams.
- Developed an autonomous robotics platform using a ESP32.
 - Implemented real-time image processing with OpenCV to localize the robot, barrels, and opponent on the playing field.
 - Designed path-planning and obstacle-avoidance algorithms for autonomous navigation.
 - Integrated wireless communication through MQTT for monitoring and control.
 - Competed in a robotics challenge involving barrel collection and delivery while avoiding collisions with opponents and obstacles.

Skills

Languages	■ French (native), Tamil (native), English (C2, TOEIC 980/990), Japanese (A1), Spanish (A1)
Programming Languages	■ Python, Java, C/C++, SQL, R, MATLAB/Octave, Dart/Flutter, SAS, VHDL
Data Science & AI	■ Machine Learning, Deep Learning, Supervised Learning, Image Processing, Reinforcement Learning, Bayesian Optimization, Signal Processing, Graph Theory
Bioinformatics	■ Single-cell RNA-seq, Transcriptomics, Gene Expression Profiling, Multi-omics Data Analysis, Tumor Microenvironment Analysis, Cell Annotation, Cell-Cell Communication, Trajectory Inference
Web Development	■ HTML, CSS, JavaScript, PHP, MySQL, Wordpress, Hostinger
Tools & Software	■ Microsoft Office (Word, Excel, PowerPoint), Microsoft Power Platform (PowerApps, Power Automate, Power BI), SharePoint, L ^A T _E X, Canva, Adobe Photoshop, SolidWorks, ThinkerCAD
Clinical & Research	■ Good Clinical Practice (ICH E6 R2), Compliance Regulations, Intellectual Property
Project Management	■ Agile / Decisioning, Consulting, Marketing, Communication
Embedded Systems	■ Embedded Linux Systems, Arduino, VHDL, Proteus, Quartus, Cisco Packet Tracer, Wireshark

Miscellaneous Experience

Awards and Achievements

- 2024  1st place out of 567 – Best Scientific Poster.
-  2nd place out of 38 – Inetum Hackathon.
- 2022  4th place out of 300 – ECE Cup Robotics Competition.

Certifications

- 2026  ICH E6 (R3) Good Clinical Practice, The Global Health Network.
- 2023  ECE Python Exam, Liora.
-  Project Management MOOC, Centrale Lille.